CHAPTER 4

Present Value, Future Value & Discounting

Discount Factors, The Temporal Equivalence Principle, Multi-Period Cash Flow Sums, and No-Arbitrage Pricing

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4.1 Definition & The Discounting Factor

The fundamental axiom of financial economics is the **Time Value of Money**: capital available today is worth more than an identical nominal amount in the future due to its earning capacity and risk premium. **Discounting** is the mathematical operation inverse to compounding, translating future obligations into today's capital.

Definition 4.1 (Discount Factors & Present Value)

Let a cash flow FV be received at time $t \geq 0$, and let $r \geq 0$ be the annual discount rate.
The **discrete discount factor** $v(t)$ and **Present Value** PV are defined by:

$$PV = FV \cdot (1 + r)^{-t} = FV / (1 + r)^{t}$$

$$PV_{\text{cont}} = FV \cdot e^{-rt} \quad \text{(under continuous discounting)}$$

where $v = (1 + r)^{-1}$ is the unit-period discount factor. Note $v(0) = 1$ and $\lim_{t \to \infty} v(t) = 0$.

4.2 The Principle of Temporal Financial Equivalence

No-Arbitrage Consistency: Two cash flow streams are economically equivalent at discount rate r if and only if their present values at time $t = 0$ are identical. Under frictionless financial markets:

- **Time Shift Invariance:** The value of capital C evaluated at time t_2 from its valuation at t_1 satisfies: $V(t_2) = V(t_1) \cdot (1 + r)^{t_2 - t_1}$.
- **Linear Additivity:** The present value of a multi-period cash flow stream $\{C_1, C_2, ..., C_m\}$ at times $\{t_1, t_2, ..., t_m\}$ is the linear sum:

Proposition 4.1 (Linear Valuation of Cash Flow Streams)

$$\boldsymbol{PV_{\text{total}} = \sum_{k=1}^{m} C_k \cdot (1 + r)^{-t_k} = \sum_{k=1}^{m} C_k \cdot v^{t_k}}$$

Under continuous discounting with term-structure rate curve $r(s)$: $PV = \sum C_k \exp(-\int_0^{t_k} r(s)\, ds)$.

4.3 Step-by-Step Worked Examples

Below are two formal case studies solved by hand: pricing a zero-coupon sovereign treasury bond, and evaluating a multi-stage deferred corporate cash flow stream.

Example 4.1: Fair Valuation of a Zero-Coupon Sovereign Bond

Problem Statement: A zero-coupon government security guarantees a bullet face value payment of ***FV* = INR 1,000,000** at maturity in ***t* = 7 years**. If the prevailing sovereign hurdle yield is ***R* = 7.25% per annum**, determine the fair theoretical price today under:

- **Case A:** Annual Discounting | **Case B:** Continuous Discounting

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Step-by-Step Analytical Solution:

Case A (Annual Discounting):

- Parameter: $r = 7.25 / 100 = 0.0725$, $t = 7$ years.
- Compounding factor: $(1 + 0.0725)^{7} = (1.0725)^{7} \approx 1.6304018$
- Discount factor: $v(7) = (1.6304018)^{-1} \approx 0.6133457$
- Fair Price: $PV_{\text{annual}} = 1{,}000{,}000 \times 0.6133457 =$ **INR 613,345.70**.

Case B (Continuous Discounting):

- Exponent: $-r \cdot t = -(0.0725 \times 7) = -0.5075$
- Discount kernel: $e^{-0.5075} \approx 0.6019979$
- Fair Price: $PV_{\text{cont}} = 1{,}000{,}000 \times 0.6019979 =$ **INR 601,997.90**.

Pricing Spread Analysis:

- Difference: $PV_{\text{annual}} - PV_{\text{cont}} = 613{,}345.70 - 601{,}997.90 =$ **INR 11,347.80**.
- *Insight:* Continuous discounting applies compounding infinitely often, creating a stronger discount drag that lowers the present fair price relative to discrete annual conventions.

Example 4.2: Multi-Stage Deferred Corporate Cash Flow Stream

Problem Statement: A corporate project promises three contractual cash inflows: $\boldsymbol{C_1}$ **= INR 200,000** at t_1 = 1 yr, $\boldsymbol{C_2}$ **= INR 400,000** at t_2 = 3 yrs, and $\boldsymbol{C_3}$ **= INR 800,000** at t_3 = 5 yrs. Using a benchmark discount rate of $\boldsymbol{R}$ **= 8.00%**, compute:

- Total Present Value (PV_{total}) at $t = 0$
- Total Future Value (FV_5) accumulated at t = 5 years

Step-by-Step Analytical Solution:

Step 1 (Discounting Individual Cash Inflows to $t = 0$):

- Year 1: $PV_1 = 200{,}000 / (1.08)^1 = 200{,}000 \times 0.9259259 =$ **INR 185,185.19**
- Year 3: $PV_2 = 400{,}000 / (1.08)^3 = 400{,}000 \times 0.7938322 =$ **INR 317,532.90**
- Year 5: $PV_3 = 800{,}000 / (1.08)^5 = 800{,}000 \times 0.6805832 =$ **INR 544,466.56**

Step 2 (Linear Summation for Aggregate Present Value):

- $PV_{total} = 185{,}185.19 + 317{,}532.90 + 544{,}466.56 =$ **INR 1,047,184.65**.

Step 3 (Compounding to Terminal Horizon $t = 5$):

By the principle of temporal financial equivalence, compounding total PV to year 5 must exactly equal compounding each individual cash flow to year 5:

- $FV_5 = PV_{total} \times (1.08)^5 = 1{,}047{,}184.65 \times 1.4693281 =$ **INR 1,538,658.00**.

Step 4 (Direct Forward Cash Flow Check):

- C_1 compounded 4 yrs: $200{,}000 \times (1.08)^4 = 200{,}000 \times 1.360489 =$ INR 272,097.80
- C_2 compounded 2 yrs: $400{,}000 \times (1.08)^2 = 400{,}000 \times 1.166400 =$ INR 466,560.00
- C_3 compounded 0 yrs: $800{,}000 \times 1.000000 =$ INR 800,000.00
- Sum: 272,097.80 + 466,560.00 + 800,000.00 = **INR 1,538,657.80** (verified exact temporal equivalence).

4.4 Algorithmic Implementation in Python

Below is the production-grade Python implementation from models/time_value.py, supporting single cash flows and multi-stage cash flow stream valuations:

Listing: models/time_value.py — Discounting & Valuation Engine

```
from typing import List, Tuple
import math

def present_value(future_value: float, annual_rate: float, time_years: float) -> float:
    """Computes discrete present value: PV = FV / (1 + r) ** t."""
    if future_value < 0.0 or annual_rate < 0.0 or time_years < 0.0:
        raise ValueError("Arguments must be non-negative.")
    return float(future_value / ((1.0 + annual_rate) ** time_years))


def present_value_stream(cash_flows: List[Tuple[float, float]], annual_rate: float) -> float:
    """
    Computes aggregate present value of stream: [(cash_flow, time_years), ...].
    PV_total = sum(C_k / (1 + r) ** t_k).
    """
    if annual_rate < 0.0:
        raise ValueError("Discount rate must be non-negative.")
    return sum(float(cf / ((1.0 + annual_rate) ** t)) for cf, t in cash_flows)
```

4.5 Mathematical Properties Summary

Mathematical Property	Specification & Formula	Analytical Significance
Discount Factor	$v(t) = (1 + r)^{-t}$	Monotonically decaying pricing kernel
Asymptotic Horizon	$\lim_{t \to \infty} v(t) = 0$	Distant cash flows have negligible present economic weight
Rate Sensitivity (Duration)	$d(PV)/dr = -t \cdot FV\ (1+r)^{-t-1}$	Negative first derivative reflects interest rate risk
Convexity	$d^2(PV)/dr^2 > 0$	Bond price convexity protects long positions against rate swings
Continuous Kernel	$v_{\text{cont}}(t) = e^{-rt}$	Standard continuous discount factor in stochastic finance